Price Discrimination

October, 2026

Where We Are

- So far every supply-side model we have written down has a firm choosing **one price per product**.

- That is a special case. A firm with market power generally chooses a **price schedule** — a map from observable consumer characteristics, or from the purchase itself, into a menu.

- Once we allow this:
  - "The" markup is no longer a scalar. Different consumers face different margins on the same good, and the *dispersion* of margins is the object of interest.
  - The welfare arithmetic is no longer "price above marginal cost." Discrimination can *expand* output and serve consumers a uniform-price monopolist would exclude.

- The empirical problem of this lecture: price dispersion is observable; discrimination is not. How to test and quantify.

Basic Concepts

- Stigler (1987): price discrimination is present when the two or more similar goods are sold at prices that are in **different ratios to marginal cost**.
  - Note the definition is about p/mc, not about p. Stigler's example: a book in hardcover vs. paperback.
  - Immediate consequence: you cannot test for price discrimination without taking a stand on marginal cost. This is the same identification problem as in the Firm Conduct lecture...

- Three conditions are necessary:
  1. **Market power**. Under perfect competition $p = mc$ for every unit sold.
  2. **Ability to sort consumers.** Either on exogenous observables (third degree) or by inducing *self-selection* through a menu (second degree).
  3. **Ability to prevent resale/arbitrage.** Trade frictions, legal restrictions, non-transferable services, or physical modification of the product.

- **Why we care**: welfare effects are theoretically **ambiguous** even in the monopoly case. That ambiguity is exactly what makes the empirical work policy-relevant.

Types of Price Discrimination

Following Varian's (1989) adaptation of Pigou (1920):

- **First-degree (perfect) PD**: the price of each unit equals the buyer's maximum willingness to pay for it. The firm extracts all surplus and the allocation is efficient.
- **Third-degree PD**: groups identified by *observed exogenous characteristics* (age, location, country, student status) are charged different *linear* prices — each group faces a constant price per unit.
- **Second-degree PD**: the menu depends on *what* the consumer buys — quantity, quality, bundle— but the same menu is offered to everyone. Often called nonlinear pricing. Pigou casts this as an "ideal" segmentation by willingness to pay; Varian's restatement emphasizes that the sorting is *endogenous*.

The information requirement is what separates them

Third degree needs the firm to **observe** the type and **enforce** the segmentation. Second degree assumes the firm observes **nothing**, and must pay *information rents* to get types to reveal themselves.

Road-map from here

- **Part I — Theory, and what it gives us to test.** First, third, and second degree; the welfare; the screening problem with a continuum of types. The point is to extract *refutable predictions*, that all the empirical papers have to work with.

- **Part II — Second degree in the field.** Shepard (1991) on full- vs. self-service gasoline: a design-based answer to "cost or discrimination?"

- **Part III — Second degree with capacity: Leslie (2004).** Broadway theater. A structural discrete-choice model with *endogenous choice sets*, and a full counterfactual comparison of PD against uniform pricing.

- **Part IV — Second degree on quality: McManus (2007).** Specialty coffee. Tests the sharpest prediction the theory has — "no distortion at the top, downward distortion below" — by measuring MB and MC of an extra ounce directly.

- **Part V — What competition does.** Armstrong–Vickers / Rochet–Stole: competition in *indirect-utility* space, and why full coverage kills the distortion.

First-Degree PD: The Benchmark

- The monopolist observes each consumer's type t and their whole demand curve, and can charge unit by unit. It sets $p = mc$ on the *last* unit and extracts all surplus through inframarginal prices — equivalently a personalized two-part tariff $T(q) = mc \cdot q + F(t)$.
- **Result**: the allocation is *first-best efficient*, consumer surplus is zero, total surplus maximized. Here monopoly distortion and distributional harm separate cleanly: **all** the harm is distributional.

- No longer only a textbook benchmark. Online firms observe rich individual data, and ML pricing is an attempt at approximate first degree.
  - Dubé and Misra (2023) run a randomized pricing experiment at ZipRecruiter. Profits rise 19% over the *optimal uniform* price, 86% over the status-quo price.
  - Aggregate consumer surplus falls 23% vs. uniform pricing — yet **over** 60% **of consumers face a lower price**. PD redistributes *within* the consumer side, not only from consumers to the firm.
  - Policy: "who is hurt" is not answerable from aggregate *CS*. Under inequity-averse welfare weights the sign can flip.

Third-Degree PD: The Pricing Rule

- Independent segments $i = 1, \ldots, N$, inverse demand $p_i(x_i)$, constant marginal cost c_i. The monopolist solves $\max_{\{x_i\}} \sum_i p_i(x_i)x_i - \sum_i c_i x_i$, which separates:

$$p_i(x_i)\left[1 - \frac{1}{\epsilon_i}\right] = c_i, \qquad \epsilon_i \equiv -\frac{\partial x_i}{\partial p_i}\frac{p_i}{x_i} > 1$$

- **Inverse-elasticity rule, segment by segment**: the Lerner index is $L_i = 1/\epsilon_i$. With common cost c, the firm charges the **higher price to the less elastic segment**.

- Note what this says nothing about which segment is richer, or more deserving. Student discounts and senior discounts are rationalized by elasticity, not by need.

- **The empirical content:** Any observed price ratio can be rationalized by *some* (ϵ_i, c_i). To make this testable you must either
  1. measure c_i independently (Shepard's strategy: find a control group with the same cost), or
  2. estimate the demand system and hence ϵ_i (Goldberg–Verboven, DellaVigna–Gentzkow, ...).

Third-Degree PD: Welfare — Varian's Bounds

- Compare uniform price p^0 (quantities x^0) to discriminatory prices p^1 (quantities x^1). Welfare is $W = u(x) - \sum_i c_i x_i$ with u concave and $\nabla u(x) = p(x)$.
- Concavity gives the two-sided bound (Varian 1985):

$$\underbrace{\sum_i (p_i^1 - c_i)\Delta x_i}_{\text{lower bound}} \;\le\; \Delta W \;\le\; \underbrace{\sum_i (p_i^0 - c_i)\Delta x_i}_{\text{upper bound}}, \qquad \Delta x_i \equiv x_i^1 - x_i^0$$

- The output test. Under a uniform price $p_i^0 = p^0$ for all i, the upper bound is $(p^0 - c)\sum_i \Delta x_i$. Since $p^0 > c$:

$$\Delta W > 0 \implies \sum_i \Delta x_i > 0$$

  Rising total output is necessary for discrimination to raise welfare. It is *not* sufficient — the lower bound can still be negative, because output may be reallocated to consumers who value it less.

- It converts an unobservable welfare question into a question about **quantities**, which we can sometimes observe directly. **Caveat**: the bounds assume the set of markets served is unchanged — if discrimination *opens* a market, the new market's surplus is a discrete addition and the test must be restated.

Third-Degree PD with Cross-Substitution

- The separability above is special: it needs demands independent across segments. If segments are substitutes — exactly in the case we studied, say for NLogit — demand is $p_i(x_i, x_j)$ and the FOCs no longer separate.
- Varian (1989) shows the ranking survives cross-substitution: if i is the larger market, $x_i > x_j$, the equality becomes an inequality.

$$\underbrace{\frac{p_i}{p_j} = \frac{1 - 1/\epsilon_j}{1 - 1/\epsilon_i}}_{\text{independent demands}} \quad \longrightarrow \quad \underbrace{\frac{p_i}{p_j} > \frac{1 - 1/\epsilon_j}{1 - 1/\epsilon_i}}_{\text{substitutes, } x_i > x_j}$$

  i.e. the price in the *large* segment is pushed *above* what the simple inverse-elasticity formula predicts.

- **Intuition**: raising p_i pushes some consumers into segment j, where the firm still earns a margin. The lost sale is partially recaptured, so the elasticity the firm *perceives* in i is lower than the own-price elasticity.
- This is precisely the mechanism Shepard exploits. A station selling *both* full- and self-service recaptures the consumers it prices out of full service; a single-product station does not.

Second-Degree PD: Setup

- Now the firm **cannot observe** the type. Two types: t_1 with population share f_1 (L) and t_2 with share f_2 (H). Quasi-linear utility $u(x,t) - r$, where x is quantity (or quality) and r is the price schedule.
- **Single-crossing**: $u(x,t_2) > u(x,t_1)$ and $\partial u(x,t_2)/\partial x > \partial u(x,t_1)/\partial x$. The H type's *marginal* willingness to pay is higher at every x. This makes a menu able to sort.
- The firm offers a menu $\{(x_1,r_1),(x_2,r_2)\}$ subject to:
- **Participation (IR)**

$$u(x_1,t_1) - r_1 \geq 0, \qquad u(x_2,t_2) - r_2 \geq 0$$

- **Self-selection (IC)**

$$u(x_1,t_1) - r_1 \geq u(x_2,t_1) - r_2 \qquad u(x_2,t_2) - r_2 \geq u(x_1,t_2) - r_1$$

- The two **bold** constraints bind at the optimum: IR_L and IC_H. The low type is held at zero surplus; the high type is just deterred from taking the low bundle.

Second-Degree PD: Why Those Two Constraints Bind

- IR_L **binds.** If $u(x_1, t_1) - r_1 > 0$, raise *both* r_1 and r_2 by ε. Profit rises, the IC constraints are unaffected (they depend on $r_1 - r_2$), and IR_H is slack because IC_H plus IR_L imply

  $$u(x_2, t_2) - r_2 \;\geq\; u(x_1, t_2) - r_1 \;>\; u(x_1, t_1) - r_1 \;\geq\; 0$$

  using single-crossing in the middle step. So IR_H is **implied**, never binding.

- IC_H **binds.** If it were slack, raise r_2 alone. Profit rises and nothing else is violated.

- IC_L **is slack** at the optimum (it holds given single-crossing and $x_2 \geq x_1$) — so we can ignore it and verify ex post.

- Substituting the two binding constraints, $r_1 = u(x_1, t_1)$ and $r_2 = u(x_2, t_2) - [u(x_1, t_2) - u(x_1, t_1)]$:

  $$\max_{x_1, x_2} \; [u(x_1, t_1) - c x_1] f_1 + \big[\underbrace{u(x_2, t_2) - c x_2}_{\text{H-type surplus}} - \underbrace{(u(x_1, t_2) - u(x_1, t_1))}_{\text{information rent}} \big] f_2$$

- The firm maximizes total surplus *minus* the rent it must concede to the H type. That rent is increasing in x_1 — the more you give the low type, the more attractive their bundle looks to the high type, and the more you must pay the high type to stay put.

Second-Degree PD: The Two Lessons

- First order conditions:

$$\underbrace{\left[\frac{\partial u(x_1, t_1)}{\partial x_1} - c\right] f_1}_{\text{efficiency gain on L}} + \underbrace{\left[\frac{\partial u(x_1, t_1)}{\partial x_1} - \frac{\partial u(x_1, t_2)}{\partial x_1}\right] f_2}_{\text{rent conceded to H } (<0)} = 0, \qquad \frac{\partial u(x_2, t_2)}{\partial x_2} - c = 0$$

- **No distortion at the top.** The H type's marginal WTP equals marginal cost: x_2 is *efficient*. Nobody is imitating the H type, so there is no rent-extraction reason to distort their quantity.
- **Downward distortion at the bottom.** By single-crossing the second bracket is negative, so the first must be positive: $\partial u(x_1, t_1)/\partial x_1 > c$. The L type consumes an **inefficiently small** amount. Shrinking x_1 destroys L-type surplus (weighted f_1) but cheapens the H type's rent (weighted f_2).
- **Comparative static**: the distortion grows with f_2/f_1. The more high types there are, the more it is worth degrading the low-end product; if f_2 is large enough the firm **excludes** the L type entirely.
- The interpretation of x is flexible: quantity, or — with unit demand — **quality**. The theory testable in coffee cups (McManus) and theater seats (Leslie).

Second-Degree PD: Continuum of Types

The two-type logic extends cleanly (Mussa and Rosen 1978). Types $\theta \sim F$ on $[\underline{\theta}, \bar{\theta}]$ with density f, utility $\theta q - T$, cost $C(q)$ convex.

- Let $v(\theta) = \max_q [\theta q - T(q)]$ be the indirect utility. The **envelope condition** replaces the entire set of global IC constraints:

$$v'(\theta) = q(\theta) \quad \Longrightarrow \quad v(\theta) = v(\underline{\theta}) + \int_{\underline{\theta}}^{\theta} q(s)\, ds$$

  with $v(\underline{\theta}) = 0$ from the binding bottom *IR*. IC additionally requires $q(\cdot)$ non-decreasing.

- Profit is $\int [\theta q(\theta) - v(\theta) - C(q(\theta))] f(\theta) d\theta$. Integrating the rent term by parts,

$$\int_{\underline{\theta}}^{\bar{\theta}} v(\theta) f(\theta) d\theta = \int_{\underline{\theta}}^{\bar{\theta}} \frac{1 - F(\theta)}{f(\theta)}\, q(\theta)\, f(\theta)\, d\theta$$

- So the firm maximizes expected **virtual surplus** pointwise:

$$\max_{q(\theta)} \underbrace{\left[\theta - \frac{1 - F(\theta)}{f(\theta)}\right]}_{\equiv \phi(\theta),\ \text{virtual type}} q(\theta) - C(q(\theta)) \qquad \Longrightarrow \qquad C'(q(\theta)) = \theta - \frac{1 - F(\theta)}{f(\theta)}$$

Reading the Continuum Solution

$$C'(q(\theta)) = \theta - \frac{1 - F(\theta)}{f(\theta)} \qquad \text{vs. the efficient rule} \qquad C'(q^{FB}(\theta)) = \theta$$

- **No distortion at the top**: $F(\bar{\theta}) = 1$, so $\phi(\bar{\theta}) = \bar{\theta}$ and $q(\bar{\theta}) = q^{FB}(\bar{\theta})$.
- **Distortion everywhere below**: $\phi(\theta) < \theta$ for $\theta < \bar{\theta}$, so $q(\theta) < q^{FB}(\theta)$. The wedge $\frac{1-F}{f}$ is the **inverse hazard rate** — the mass of types above θ who must be bribed, per unit of density at θ.
- **Exclusion at the bottom**: if $\phi(\underline{\theta}) < C'(0)$, low types get $q = 0$. Nonlinear pricing generically leaves part of the market unserved.
- **Regularity**: ϕ increasing (monotone hazard rate) guarantees $q(\theta)$ non-decreasing, hence IC. Otherwise the solution is "ironed" — flat segments, i.e. **bunching**, seen empirically as a coarse menu with few items.

The two testable predictions

(1) The top item in a product line is priced at marginal cost *on the margin* — its quality/size is efficient. **(2)** Every item below it is distorted **downward**, and the distortion is larger the further down the line you go.

What the Theory Predicts

Strategy	Prediction tested	What must be assumed	Paper
Design-based: find a cost control	$\Delta_f \geq 0$, $\Delta_s \leq 0$ across station types	costs do not vary with retail configuration	Shepard (1991)
Structural demand, then simulate	PD vs. uniform pricing: ΔCS, $\Delta\pi$	demand system; firm optimizes; capacity rationing rule	Leslie (2004)
Structural demand + direct cost data	$MB(q) = MC$ at the top, $MB(q) > MC$ below	utility functional form; engineering *mc* per ounce	McManus (2007)

- The same trade-off as in the Monopoly Power lecture: Shepard needs almost no structure but can only sign a differential; Leslie and McManus get magnitudes and welfare, at the price of a demand system.

Shepard 1991: The Identification Problem

- One of the first papers to ask whether observed price dispersion across seemingly similar goods is **discrimination** rather than **cost**.
- Setting: retail gasoline, eastern Massachusetts, a 12-week period in early 1987; roughly 1,000–1,300 stations per grade.
- The confound: full service costs more to supply than self service (an attendant pumps it). So $p_f > p_s$ proves nothing.
- The design: compare the full-/self-service price gap at stations selling **both** (multiproduct, MP) with the gap implied by stations selling **only one** (single-product, SP).
  - Treatment: MP stations, which *can* discriminate because they offer a menu. Control: stand-alone full- and self-service stations, which cannot.
  - **Key identifying assumption**: the cost of a given service does not depend on whether the station also offers the other one.
- **The logic**: only MP stations face the self-selection problem of Part I, so the SP configuration gives the counterfactual price *absent* the ability to sort. The dif-in-dif across configuration and service type is the discrimination effect.

Shepard: A Simple Model

- Abstract from horizontal (spatial) differentiation; treat each station as a local monopolist. Service type $g \in \{f, s\}$, configuration $k \in \{MP, SP\}$.
- Consumer type $t \sim U[0,1]$, with utility

$$U = \begin{cases} V(g)\,(t - p_g) & \text{if one unit of service } g \text{ is bought} \\ V(0)\,t & \text{if no purchase} \end{cases}$$

  where $V(f) > V(s) > V(0) > 0$. Higher t means higher value, and $V(g)$ scales *both* the value of the good and price sensitivity — delivers single-crossing here.

- **Cutoffs**, derived by equating utilities:

$$\text{buy } g \text{ vs. not:} \quad V(g)(t - p_g) = V(0)t \implies t_{g0} = \frac{V(g)\,p_g}{V(g) - V(0)}$$

$$f \text{ vs. } s: \quad V(f)(t - p_f) = V(s)(t - p_s) \implies t_{fs} = \frac{V(f)p_f - V(s)p_s}{V(f) - V(s)}$$

- So at an MP station consumers sort: $t > t_{fs}$ buy full service, $t_{s0} < t < t_{fs}$ buy self-service, the rest do not buy. This is second-degree PD — the station posts a two-item menu and lets consumers reveal t.

Shepard: Demands

- Single-product station offering only g ($D = 1 - t_{g0}$):

$$D(p_g) = 1 - \frac{V(g)\, p_g}{V(g) - V(0)}$$

- Multiproduct station (using $D_f = 1 - t_{fs}$ and $D_s = t_{fs} - t_{s0}$):

$$D_f(p_f, p_s) = 1 - \frac{V(f)p_f}{V(f) - V(s)} + \frac{V(s)p_s}{V(f) - V(s)}$$

$$D_s(p_s, p_f) = \frac{V(f)p_f}{V(f) - V(s)} - \frac{V(s)[V(f) - V(0)]\, p_s}{[V(f) - V(s)][V(s) - V(0)]}$$

- For the SP station the relevant own-price elasticity is governed by $V(g) - V(0)$ — the comparison is with *not buying*. For the MP station the full-service demand is governed by $V(f) - V(s)$ — the comparison is with *switching to self-service*.

- Since $V(f) - V(s) < V(f) - V(0)$, the MP station's full-service demand is **less** elastic at the relevant margin: a consumer priced out of full service is not lost, merely downgraded. This is the cross-substitution recapture effect from the Varian (1989) slide.

Shepard: Optimal Prices

Let short-run marginal costs be w for self-service and $w + \alpha$ for full-service ($\alpha > 0$ is the attendant cost).

- **Single-product** (standard linear-demand monopoly, "half the choke price plus half the cost"):

$$p_f^{SP} = \frac{V(f) - V(0)}{2V(f)} + \frac{w + \alpha}{2}, \qquad p_s^{SP} = \frac{V(s) - V(0)}{2V(s)} + \frac{w}{2}$$

- **Multiproduct**: the two FOCs, after clearing denominators, reduce to a linear system in (p_f^{MP}, p_s^{MP}). Write $A = V(f) - V(s)$, $B = V(s) - V(0)$, $C = V(f) - V(0)$, $S = V(f) + V(s)$:

$$\begin{aligned} 2V(f)\,p_f \;-\; S\,p_s \;&=\; A \;+\; V(f)\,\alpha \\ B\,S\,p_f \;-\; 2V(s)\,C\,p_s \;&=\; V(s)\,B\,\alpha \;-\; V(s)\,A\,w \end{aligned}$$

- Note that w drops out of the *first* FOC entirely: the full-service margin at an MP station is disciplined by the *self-service price*, not by the common cost component.

Shepard: The Testable Prediction

- On the region of the (w, α) space where all service/station types have positive sales:

$$\Delta_f \equiv p_f^{MP} - p_f^{SP} \geq 0, \qquad \Delta_s \equiv p_s^{MP} - p_s^{SP} \leq 0, \qquad \Delta \equiv \Delta_f - \Delta_s > 0$$

- **Why** $\Delta_f \geq 0$: an MP station raising p_f loses its marginal full-service customer only to its *own* self-service pump, where it still earns a margin. An SP station loses the sale outright. More scope to mark up.
- **Why** $\Delta_s \leq 0$: cutting p_s attracts new customers *and* pulls marginal customers down from full service. For the SP station only the first effect exists, and it is already optimized.
- Δ is the object of interest, so it is a difference-in-differences: it nets out any level effect common to MP stations (their being bigger, or on busier roads).

Shepard: Empirical Specification

- **The threat**: economies of scope. If offering both services is cheaper per unit than offering one, MP stations have lower costs and should have lower prices for *both* services.

- Shepard's argument: such scope economies plausibly lower p_s^{MP} more than p_f^{MP}, which raises Δ_s but also *shrinks* Δ_f. The net effect on Δ is to bias **against** finding discrimination. So a positive estimate of Δ is conservative.

- Specification, run separately by gasoline grade:

$$p_{ikgj} = \beta_0 + \beta_1 D_g + \beta_2 D_k + \beta_3 D_k D_g + \phi X_{ikg} + \epsilon_{ijkg}$$

  with i station, j market, $k \in \{MP, SP\}$, $g \in \{f, s\}$; $D_g = 1$ for full service, $D_k = 1$ for multiproduct.

- Mapping to the theory: $\beta_2 = \Delta_s$, $\beta_2 + \beta_3 = \Delta_f$, $\boxed{\beta_3 = \Delta}$.
- X controls for station attributes that shift cost: brand, number of pumps/capacity, repair bays, convenience store, station age, market fixed effects.

Shepard: Results

TABLE 2

PRICE DIFFERENTIALS BY GRADE

	Regular Leaded	Regular Unleaded	Premium Unleaded
Constant	75.47	83.02	97.18
	(1.36)	(1.48)	(1.59)
D_g ($\bar{\Delta}_{SP}$)	6.89	7.64	8.04
	(1.45)	(1.56)	(1.68)
D_n ($\bar{\Delta}_s$)	.00	−2.89	−2.03
	(1.67)	(1.79)	(1.90)
D_gD_n ($\bar{\Delta}$)	9.39	11.23	9.22
	(1.58)	(1.69)	(1.82)
UNBRANDED	−1.97	−4.65	−6.44
	(.55)	(.53)	(.58)
MINI	.19	2.96	2.88
	(.90)	(1.01)	(1.07)
SPFCAP	−.89	−.72	−.70
	(.16)	(.16)	(.17)
SPSCAP	−.21	−.28	−.17
	(.18)	(.20)	(.21)
MPCAP	−.21	.25	.16
	(.18)	(.18)	(.19)
REPAIR	1.80	.38	.11
	(.55)	(.59)	(.63)
CSTORE	1.43	.68	−.57
	(.70)	(.76)	(.81)
NEW	−1.40	−1.66	−1.64
	(.39)	(.41)	(.44)
STATIONS	1,052	1,291	1,237
R^2	.46	.45	.42

NOTE.—Standard errors are in parentheses.

Prices in cents/gallon. $D_g = \bar{\Delta}_{SP}$, $D_n = \bar{\Delta}_s$, $D_g D_n = \bar{\Delta}$. Standard errors in parentheses.

Shepard: What to Take Away

- A clean demonstration that a sizeable share of observed price dispersion in a near-commodity retail market is discrimination, obtained with almost no structure.
- Find two sellers facing the same costs but differing in their *ability to sort*, and difference.
- The main concern: **retail configuration is a choice**. Stations that expect to gain most from discrimination — those with the most heterogeneous customers — are the ones that become MP. That selection would bias $\hat{\Delta}$ **upward**. Shepard controls for observables, but the sorting is on unobservables; later work treats product configuration as a selection problem explicitly.
- Also note what the design *cannot* show: no welfare statement, because we never learn the demand system. We can say discrimination *exists* and is *large*, but not whether it helped or hurt.

Leslie 2004: Price Discrimination in Broadway Theater

- Setting: the 1996 Broadway run of *Seven Guitars*, with performance-level data on prices, attendance, and revenue by ticket category. Both forms of discrimination operate at once:
  - **Second degree**: a menu of seat *qualities* (orchestra, mezzanine, balcony); plus the TKTS **discount booth**, selling day-of-performance tickets at 50% off *if you queue*.
  - **Third degree**: **targeted coupons** to identified groups (students, subscribers, direct mail).
- Why the booth is the interesting instrument: queuing is a costly, *self-selecting* screen whose cost is higher for high-opportunity-cost consumers — exactly the sorting device second-degree PD needs.
- The methodological contribution is a discrete-choice model with **endogenously random choice sets**: capacity means your preferred option may be sold out, and coupons mean some options are never offered to you.
- Payoff: counterfactuals comparing the observed scheme with **uniform pricing** and with variants of the booth.

Leslie: The Price Variation to Be Explained

- Price categories (orchestra, front/rear mezzanine, balcony, boxes, standing room)
- Peak-load variation (Saturday evening vs. Thursday evening, etc.)
- Discounts: TKTS booth, two-for-one coupons, student, direct mail

TABLE 1 **Summary of Attendance and Revenues for Each Sales Category of *Seven Guitars***

	Price ($)		Attendance		Revenue ($)	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Full price						
Orchestra	55.08	4.22	162.74	77.22	9,112.29	4,765.14
Front mezzanine	55.08	4.23	40.04	41.70	2,262.27	2,462.55
Rear mezzanine	29.20	1.85	34.80	18.91	1,007.10	533.49
Balcony	16.93	4.91	38.60	17.26	679.26	421.85
Boxes	55.76	4.17	4.97	4.88	281.36	279.95
Standing room	22.27	2.55	6.14	4.50	134.77	96.24
Discount price						
10% off	49.40	3.88	6.71	5.55	335.74	286.61
Two-fer one	27.23	2.06	16.65	20.17	467.28	591.53
TKTS	27.53	2.11	158.87	71.29	4,358.12	1,956.91
MTC	22.00	0	258.99	60.28	5,697.71	1,326.18
AENY	50.36	1.81	3.81	2.46	193.07	128.17
Direct mail	39.51	2.28	48.43	36.80	1,925.78	1,461.92
Group	36.26	10.80	89.91	63.84	3,309.46	2,688.23
Student	26.21	2.01	68.35	56.38	1,775.98	1,440.68
TDF	16.46	5.81	153.72	90.67	2,306.93	1,163.35
Wheelchair	26.94	2.23	2.02	0.66	54.56	17.67
Complimentary	0	0	38.91	75.57	0	0

Notes: "Two-fer one" are two-for-one coupon sales. "TKTS" are tickets sold via the day-of-performance discount booths. "MTC" stands for Manhattan Theatre Club, which is a subscriber organization. "AENY" stands for Arts Entertainment New York, which is a private firm specializing in providing high-quality tickets to its customers. "TDF" stands for Theater Development Fund, which is a nonprofit organization that provides tickets to schoolchildren and so forth. In the model, "coupons" are the aggregation of all discount-price categories except TKTS, wheelchair, and complimentary tickets.

Orchestra averages \$55.08; TKTS \$27.53 — a 50% discount that moves 159 seats a performance, nearly as many as the full-price orchestra (163).

Leslie: Modeling Choices

- Consumer heterogeneity is two-dimensional: (y_i, ξ_i) = income and taste for theater, distributed F and G. Crucially, neither is observed — all cross-sectional variation in shares must be explained by the distribution of (y, ξ).
- Subject to *availability*, utility from a full-price ticket of quality $j \in \{\ell, m, h\}$, given budget $B(y_i)$:

  $$U_{ij} = q_{ij}\,[B(y_i) - p_j]^{\eta}, \qquad B(y_i) = \delta_1 y_i^{\delta_2}$$

  The **multiplicative** form in quality is what generates sorting: the marginal value of quality rises with residual income $B(y_i) - p_j$, so richer consumers self-select into better seats. (Single-crossing, in the Part I language.)
- With probability $\lambda(y_i \mid \gamma)$ — the firm's *targeting technology* — the consumer receives a coupon price $p_c < p_h$ for an orchestra seat.
- The booth is modeled with an explicit **queuing cost** that rises with income:

  $$U_{ib} = q_{ib}\big[B(y_i) - p_b - (\tau_1 y_i + \tau_2)\big]^{\eta}$$

- Outside option: $U_{i0} = \xi_i^{-1}[B(y_i) - p_0]^{\eta_0}$, with $p_0 = 0$ and $\eta_0 = 1$ normalized — so a *high* ξ_i means a *low* outside option, i.e. a strong taste for attending tonight.

Leslie: Capacity, Rationing, and Random Choice Sets

- Each quality segment j has a hard capacity C_j (the house has a fixed number of orchestra seats). Leslie simulates a **sequence** of N_t arriving consumers (y_i, ξ_i), each choosing from what is still available; when C_j is exhausted, option j is removed for everyone after. Averaged over 1,000 random arrival permutations.
- If a consumer does not draw a coupon, option c is simply not in their choice set. So **choice sets differ across consumers** for two distinct reasons — rationing (endogenous, depends on prices) and targeting (exogenous to the consumer).
- With binding capacity, a price cut need not raise attendance, and the observed share of a sold-out category is a *censored* measure of demand. Ignoring this would attribute the censoring to low willingness to pay and badly bias the elasticities — and hence the counterfactuals.
- Conditional on the attendance shifters ξ_{it}, all variation in the purchase shares s_{jt} *across categories* is driven by the unobserved income distribution.
- Parametrization: $\xi_{it} \sim \text{Exp}(X_t\beta)$ with X_t containing day-of-week dummies, advertising (28-day moving average), Tony Awards, number of competing shows, and a time trend; $\lambda_{it} = [1 + \exp(\alpha y_i - Z_t\gamma)]^{-1}$.

Leslie: Estimation and Identification

- At each t, for each simulated consumer, construct the choice region

  $$A_{jt} = \{(y, \xi) : U_{ijt} \geq U_{ikt} \ \ \forall k \in (\text{available set})\}$$

  and integrate to get the simulated share

  $$s_{jt}(p_t, X_t, Z_t, \Theta) = \int_{A_{jt}} dF(y)\, dG(\xi \mid X_t \beta)$$

- Estimated by **simulated maximum likelihood** on 4.9 million ticket-level observations. Similar in spirit to BLP, but with **no unobserved product-quality shock** and hence **no contraction-mapping inversion** — the model is simulated forward and matched directly.

- Identification:
  - Prices are argued exogenous conditional on quality: the grid was largely set in advance and rarely adjusted within the run. The coupon calendar was likewise preset, so it does not respond to demand shocks at t.
  - **Beyond that, identification is functional form.** (y_i, ξ_i) are never observed; the income distribution is pinned down only by requiring the *same* F to rationalize category shares at every performance. This is the paper's weakest link.

Leslie: Parameter Estimates

TABLE 3 Estimated Parameters

	q_m	1.6921	(.0064)
	Q_{max}	3.3314	(.0244)
	δ_1	2.5199	(.0163)
	δ_2	.4414	(.0007)
	τ_1	.0067	(.0000)
	τ_2	2.7365	(.0305)
	η	1.0316	(.0022)
β:	Constant	.0180	(.0006)
	Advertising ($'00,000)	.0100	(.0005)
	Tony Awards	.0008	(.0002)
	Saturday evening	.0307	(.0015)
	Friday evening	.0080	(.0010)
	Sunday evening	.0237	(.0038)
	Sunday matinee	.0045	(.0016)
	Saturday matinee	.0040	(.0004)
	Thursday evening	.0050	(.0011)
	Number of other shows	.0094	(.0001)
	$t/100$	.0525	(.0008)
γ:	Constant	21.2021	(.1045)
	Manhattan Theatre Club	−.8105	(.0406)
	Saturday evening	−3.1797	(.1322)
	Friday evening	−1.9682	(.1144)
	Sunday evening	.6080	(.6669)
	Sunday matinee	−.2090	(.0879)
	Saturday matinee	−.1995	(.0820)
	Thursday evening	−.3824	(.1284)
	$t/100$	−4.4849	(.0635)
	$t^2/10,000$	−.0653	(.0076)
	Number of observations		4,886,572
	Log-likelihood		−776,703.44

Notes: Standard errors are in parentheses. The following normalizations were applied: $q_t = 1$, $\eta_o = 1$, $p_o = 0$, and $\alpha = .01$. Advertising is a moving average over the previous 28 days.

Normalizations $q_\ell = 1$, $\eta_0 = 1$, $p_0 = 0$, $\alpha = 0.01$. Quality ladder $\hat{q}_m = 1.69$, best seat $\hat{Q}_{\max} = 3.33$ — a threefold spread. $\hat{\eta} = 1.03$. Queuing cost $\hat{\tau}_1 = 0.0067 > 0$: the booth's time cost rises with income, which is what makes it a screen rather than a giveaway.

Leslie: Counterfactual Design

- Supply side: the firm chooses $p = \{p_\ell, p_m, p_h, p_b, p_c\}$ to maximize **revenue** — the marginal cost of seating one more person in a theater that is running anyway is essentially zero.
- So **total surplus** $\propto$ **revenue** + **consumer utility**: revenue and consumer welfare need *not* move in opposite directions.
- **Base-A** = model at *observed* prices. **Base-B** = model at *optimal* prices under the existing PD structure (including the mandated 50% booth discount).
- Base-A revenue is \$6.27m against \$4.70m realized, attendance 907 vs. actual 662 — the model over-predicts levels, largely because many discount categories are collapsed into one "coupon" price. **All comparisons below are model-vs-model**, with Base-A as baseline.
- The experiments: **Uniform** (one price, no booth, no coupons); **No-booth-A/B** (remove the booth, holding other prices fixed / re-optimizing); **Booth not 50%** (keep the booth, free the discount); **Nonsticky** (prices vary by performance).

Leslie: Counterfactual Results

TABLE 5 **Results of Counterfactual Experiments**

Experiment	Revenue ($ million)	Utility	Average Attendance	p_ℓ	p_m	p_h	p_b	p_c
Actual	4.6951	NA	661.56	16.93	29.20	55.08	27.53	31.01
Base-A	6.2698	3.5859	906.86	16.93	29.20	55.08	27.53	31.01
Base-B	7.8965	3.5775	864.11	23.90	29.80	60.22	30.11	45.26
Uniform	8.0204	3.6039	809.57	50.04	50.04	50.04	NA	NA
No-booth-A	6.7301	3.5837	873.01	16.93	29.20	55.08	NA	31.01
No-booth-B	8.3495	3.5925	873.73	22.28	38.33	51.53	NA	43.23
Booth not 50%	8.4516	3.5900	850.30	24.47	40.86	54.21	38.05	46.32
Nonsticky	8.0194	3.5800	887.37	24.11	30.11	59.73	29.87	46.03

Notes: See Section 5 for explanations of each experiment. The prices shown are the average prices across all performances. For some experiments, prices do not change from performance to performance, for others they do. The figure for average actual attendance does not include wheelchair tickets, standing room, and complimentary tickets. If these categories are included, the average actual attendance is 707.

Leslie (2004), Table 5. Revenue in $m over the run; "Utility" is average consumer utility; prices are averages across performances.

Leslie: Counterfactuals

- **Optimizing prices matters a lot**: Base-A $\rightarrow$ Base-B raises revenue from \$6.27m to \$7.90m (+26%), while average utility is roughly flat (3.586 $\rightarrow$ 3.578).
- The headline surprise: **Uniform pricing** delivers *both* higher revenue (\$8.02m vs. \$7.90m) *and* higher consumer utility (3.604 vs. 3.578) than optimized price discrimination. On these estimates the observed PD scheme is dominated.
- **Why?** The culprit is the 50% booth discount, which is an industry convention rather than an optimized price. Free it and PD wins on revenue again:
  - *Booth not 50%*: revenue \$8.45m — the highest of any experiment — with utility 3.590.
  - *No-booth-B*: revenue \$8.35m, utility 3.593. Simply abolishing the booth and re-optimizing beats uniform pricing on revenue.
- **So the standard theoretical model works after all**: unconstrained second-degree PD raises firm revenue (\$8.45m vs. \$8.02m, +5%) and lowers consumer utility (3.590 vs. 3.604, −0.4%) relative to uniform pricing. Both effects are modest.

Leslie: What to Take Away

- The welfare cost of price discrimination here is **small** — a fraction of a percent of consumer utility — and attendance is similar across regimes.
  - Zero marginal cost plus binding capacity means there is little allocative efficiency at stake. PD is almost purely a **transfer**.

- The most striking finding is that a real, long-standing pricing institution — the 50% TKTS rule — leaves money on the table.

- Methodologically, the contribution is the **endogenous choice set**: capacity-constrained discrete choice is now standard in airlines, hotels, and event pricing.

- **Caution**: this is one show, one theater, a revenue-maximizing objective, and a model that over-predicts attendance by 37%. The *sign* of the comparisons is more credible than their magnitude.

McManus 2007: Testing the Theory

- Recall the two sharp predictions of second-degree PD from Part I: **no distortion at the top** and **downward distortion below**. Leslie's counterfactuals do not test them — they are about welfare, not the shape of the menu.
- McManus tests them directly, where the "quality" dimension is **physically measurable**: the **size** of a cup of coffee. Setting: specialty coffee around the University of Virginia — an **oligopoly** of several shops (Greenberry's, Starbucks, Espresso Corner, Espresso Royale, campus cafeterias), each posting a menu of sizes.

- Why this setting is well chosen:
  1. Size is one-dimensional and observed in ounces.
  2. The marginal cost of an extra ounce can be **computed from engineering data** — recipes, ingredient prices, labor schedules — rather than inferred from a pricing model.
  3. Several firms with different menus, so we can ask whether competition disciplines the distortion.
- **Finding (preview)**: product sizes are close to efficient for the largest, most expensive drinks; smaller drinks are systematically **too small**.

McManus: Demand Model

- A period t is a time window of the day (morning / afternoon / night / weekend); N_t potential consumers choose from the products $j \in J_t$ available then, or the outside option. Three categories x: d (drip), r (regular espresso), s (sweet espresso), each in several sizes.
- Consumers differ by location l (dorm, classroom building, etc.):

$$U_{ijlt} = \beta_{xi}\, q_j^{\gamma_x} + \alpha p_j + \delta D_{jl} + \xi_j + \epsilon_{ijlt}$$

  where q_j is size in ounces, D_{jl} is distance from l to the store selling j, ξ_j is a product fixed effect, and the random coefficient is $\beta_{xi} = \beta_x \exp(\sigma_x \nu_{xi})$ with $\nu_{xi} \sim N(0,1)$.

- Two modeling choices:
  - $q_j^{\gamma_x}$ with $\gamma_x < 1$: utility is **concave** in size, so an interior optimal size exists. γ_x is estimated, not imposed.
  - β_{xi} log-normal and heterogeneous: this *is* the **type distribution** $F(\theta)$ of the screening model. With no heterogeneity in the taste for size there is nothing to screen on.
- Shares integrate over tastes, shocks, and locations: $s_{jt}(\theta) = \int_{A_{jt}} dF_\epsilon(\epsilon)\, dF_\nu(\nu)\, dF_D(D; t)$.

McManus: Estimation

- Maximum simulated likelihood on the observed purchase counts n_{jt}:

$$L(\theta) = \sum_{t=1}^{T} \sum_{j \in J_t} n_{jt} \log\left(s_{jt}(\theta)\right)$$

- Why MSL rather than BLP-style GMM: the number of products j is small while the panel T is long, so the product effects ξ_j can be **estimated directly as parameters** instead of inverted out — no instruments needed for ξ, because no large-market unobservable is left over. (The price: ξ_j is constant over t, so any *time-varying* unobserved quality still biases $\hat{\alpha}$ toward zero.)

- Aggregate over locations with observed time-varying weights ω_{lt} (where people are on campus at 8am vs. 10pm), then integrate the logit share over the random coefficient (ϵ i.i.d. type-I extreme value):

$$s_{jt}(\theta) = \sum_{l=1}^{L} \omega_{lt}\, s_{jlt}(\theta), \qquad s_{jlt}(\theta) = \int \left[\frac{\exp\left(\delta_{ijlt}(\nu)\right)}{\sum_{j' \in J_t} \exp\left(\delta_{ij'lt}(\nu)\right)} \right] dF_\nu$$

McManus: Estimates

TABLE 5 Utility Parameter Estimates

Price (α):	−4.509	Distance (δ):	−1.802
	(.064)		(.081)
	Drip Coffee	Regular Espresso	Sweet Espresso
Median utility (β)	.388	.084	1.213
	(.050)	(.009)	(.107)
Preference variation (σ)	1.643	2.636	1.260
	(.072)	(.056)	(.046)
Utility curvature (γ)	.415	.360	.334
	(.071)	(.014)	(.008)
Fixed Effects (ξ)			
GB—Alderman	−1.049	−.915	−1.108
	(.142)	(.175)	(.210)
GB—Bookstore	−1.573	−2.021	−1.494
	(.088)	(.087)	(.200)
GB—Poolside	−3.260	N/A	N/A
	(.058)		
GB—Tuttle	−2.416	−.720	−3.277
	(.032)	(.303)	(.243)
Espresso Corner	2.233	1.547	2.718
	(.154)	(.107)	(.120)
Starbucks	1.963	3.281	3.018
	(.095)	(.161)	(.138)
Espresso Royale	1.686	2.011	2.059
	(.043)	(.097)	(.309)
HG—East Cafeteria	1.813	−.796	.299
	(.115)	(.292)	(.146)
HG—West Cafeteria	*.603*	*−2.387*	*−2.215*
	(.137)	*(.121)*	*(.417)*
	Afternoon	Night	Weekend
Outside Options (ϕ)	1.650	.651	−.926
	(.177)	(.137)	(.252)

Note: Standard Errors are in parentheses. Abbreviations: GB = Greenberry's, HG = Higher Grounds. Greenberry's Poolside does not offer regular or sweet espresso drinks.

$\hat{\alpha} = -4.509$ (price), $\hat{\delta} = -1.802$ (distance). Curvature $\hat{\gamma}_x = 0.42, 0.36, 0.33$ — well below 1, so utility is concave in size. Dispersion $\hat{\sigma}_x = 1.64, 2.64, 1.26$: large heterogeneity in the taste for size, the precondition for screening to pay.

McManus: Measuring the Distortion

- From the utility function, the **marginal benefit of an extra ounce**, converted to dollars by the marginal utility of income $|\alpha|$:

$$MB_{ijlt}(\theta) = \frac{\gamma_x \, \beta_{xi} \, q_j^{\gamma_x - 1}}{|\alpha|}$$

  averaged over consumers i and locations l who actually choose j. Because $\gamma_x < 1$, MB **falls with size** within a product line — that is the within-line gradient the theory predicts something about.

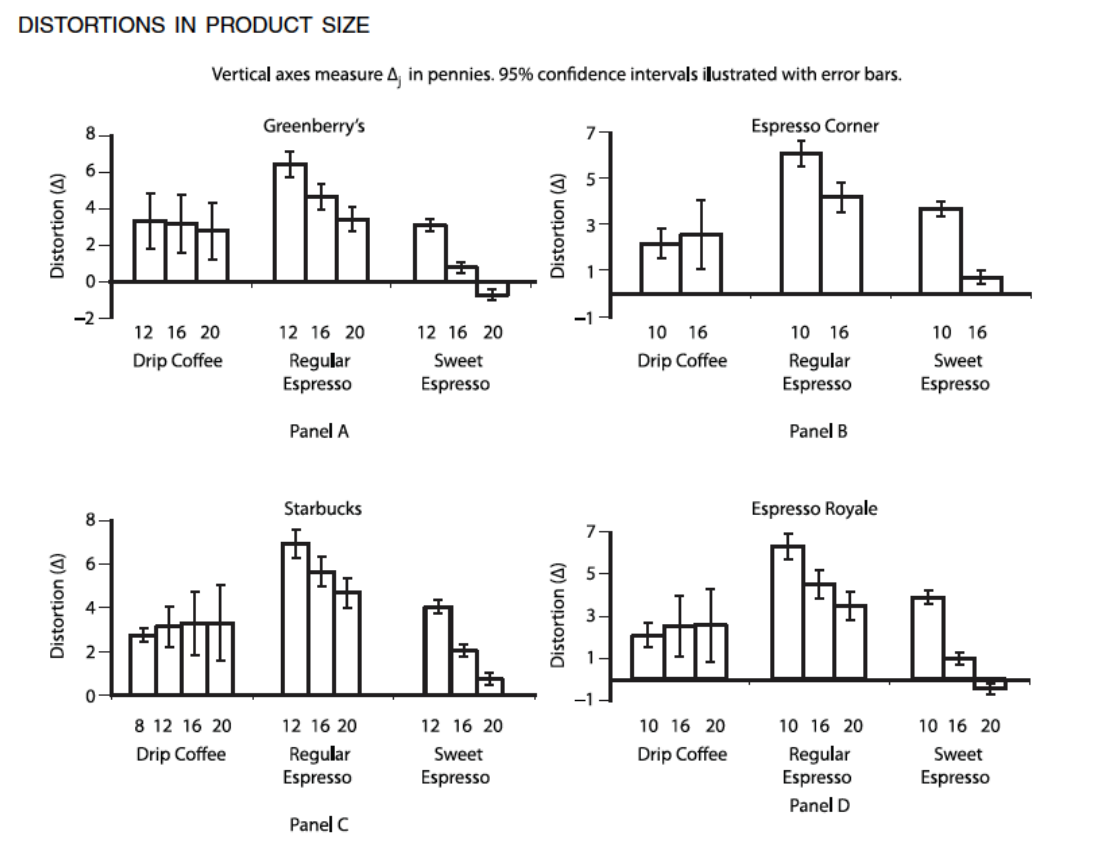
- Compare with the marginal cost of an extra ounce, built from **direct cost data**: labor schedules, drink recipes, material expenses.
  - **Key accounting insight**: the labor of making a drink is incurred regardless of size, so the marginal cost of an ounce is *only* the additional ingredients.
  - mc per ounce: drip coffee \$0.02, regular espresso \$0.04, sweet espresso \$0.06.

- The test statistic is the **distortion**, in pennies per ounce:

$$\Delta_j \;=\; \overline{MB}_j - mc_{x(j)}$$

- Predictions: $\Delta_j = 0$ for the top item in each line (no distortion at the top); $\Delta_j > 0$ below it (sizes too small); and Δ_j **declining in size** within a line.

McManus: Results



McManus: Results

- **The within-line gradient is clean.** For regular espresso at Greenberry's, Δ falls from about 6.4 to 4.7 to 3.4 cents as size goes $12 \to 16 \to 20$ oz; the same monotone pattern appears at Starbucks, Espresso Corner, and Espresso Royale. Sizes are too small, and *progressively less so* as you move up the line.
- **"No distortion at the top" holds where it should bite hardest.** Sweet espresso is the most expensive line, and its largest size has Δ statistically indistinguishable from zero (slightly negative at Greenberry's and Espresso Royale). The top of the top line is efficiently sized.
- **Magnitudes are economically large at the bottom.** A 12 oz regular espresso has $MB \approx 6$–7 cents/oz against a marginal cost of 4 cents/oz — marginal benefit is over $1.5\times$ marginal cost. The small drink is substantially degraded.
- Where the theory does less well: **drip coffee** shows no gradient at all — Δ is flat around 2–3 cents across 8–20 oz and imprecisely estimated. Plausibly drip coffee is close to a competitive, undifferentiated fringe product where there is little to screen.

McManus: What to Take Away

- This is the most direct available test of the Mussa–Rosen predictions, and they survive — in an *oligopoly*, not a monopoly. Competition among four or five nearby shops does not eliminate quality degradation.

- That is informative for Part V: the Armstrong–Vickers / Rochet–Stole benchmark says competition *with full market coverage* removes the distortion entirely. Finding distortions here means full coverage **fails** — the outside option (no coffee) binds, and firms still have a participation margin to exploit.

- The design's real innovation is measuring *mc* **outside** the pricing model. Everywhere else in this lecture, *mc* is inferred from the very optimality conditions being tested — so a rejection could always be blamed on the cost model. Here it cannot.

- **Limits**: the engineering *mc* ignores any size-related spillage, cup, or lid cost, which would bias *mc* down and the measured distortion up. And the test is about the *menu*, not about welfare — it tells us sizes are distorted, not how much surplus that destroys.

A Caution: The Distortion Can Go the Other Way

- Mussa–Rosen says quality is distorted **downward**. That rests on the monopolist facing a *fixed* outside option and screening a given set of participants.
- **Crawford, Shcherbakov, and Shum (2019)** study US cable TV, 1997–2006, estimating demand and quality (channel count) jointly, and find the opposite:
  - Prices are 33–74% **above** and qualities 23–55% **above** the socially optimal levels.
  - Customers would prefer **smaller, lower-quality bundles at a lower price**; at the social optimum the average consumer's surplus roughly **doubles**.
- The mechanism: the pressure comes from high-end **satellite** rivals. To retain high-value subscribers who could defect, cable operators load up the bundle — and then charge everyone for it. Absent satellite, the same firms would **degrade** quality in the familiar way.
- **The general lesson**: the sign of the distortion is set by *who the marginal consumer is*. Screening against an inside rival at the top pushes quality up; screening against an outside option at the bottom pushes it down.

Third-Degree PD: The Empirical Agenda

- **International price discrimination.** Goldberg and Verboven (2001, 2004) document large and persistent dispersion in auto prices across European countries — the same car model costing substantially more in one country than another. Three candidate explanations:
  1. different elasticities, hence different optimal markups;
  2. different costs (transport, right-hand drive, dealer margins);
  3. trade barriers — import quotas, exclusive dealing, grey-market restrictions.
- Using a multiproduct oligopoly model with differentiated products, they find **(1) does most of the work**: there is a strong *home-brand bias* in demand (Italian consumers' preference for Fiat, French for Renault/Peugeot), which hands domestic producers a less elastic home segment and a higher markup. Goldberg and Verboven (2005) find convergence after market integration, but it is slow and incomplete.

- Note the method: there is no cost control group here. Identification comes entirely from estimating the demand system, so the markup is whatever the elasticities say it is after costs are parameterized. Contrast Shepard.

The Opposite Puzzle: Uniform Pricing

- DellaVigna and Gentzkow (2019) find that US food, drugstore, and mass-merchandise chains charge **nearly uniform prices across their own stores**, despite wide variation in local demographics and competition.
- Their demand estimates imply substantial within-chain variation in elasticities, so this is *not* optimal: the median chain forgoes about **$16m of annual profit** relative to store-level optimal prices.
- Differences in *average* prices *across* chains *are* broadly consistent with the optimal benchmark — firms are not ignoring elasticities altogether, they just do not act on *within*-chain variation.
- The explanations the authors favor are not in our model at all: **managerial inertia** and **brand-image / fairness concerns**, both volunteered by industry participants.
- Distributional sting: optimal store-level pricing means *lower* prices in poorer, more elastic areas, so uniform pricing raises prices in poor areas relative to the discriminatory benchmark — **banning third-degree PD is not automatically progressive**.

Part V: Introducing Competition

- Everything in Part I was monopoly screening (Mussa–Rosen 1978): a single firm offers a menu $(q(\theta), P(\theta))$ to types θ.

- But McManus's coffee shops and Shepard's gas stations are **oligopolists**. What happens when several firms simultaneously offer nonlinear tariffs? The theory is surveyed in Stole (2007), and is stylized.

- The core difficulty: with competition, a consumer's outside option is no longer a constant — it is the *best rival menu*, which is itself type-dependent and endogenous. The participation constraint becomes a type-by-type object, so the standard "bind at the bottom" logic collapses.

- **Key idea that restores tractability**: competition takes place in the space of **indirect utility** $v(\theta)$ rather than in the space of prices. Firms compete by offering *utility levels* to each type, and the menu is recovered afterwards.

Model Setup

- Consumers: type θ distributed on $[\underline{\theta}, \bar{\theta}]$ with pdf $f(\theta)$.
- Duopoly: firm $j \in \{A, B\}$ offers a nonlinear price schedule $P_j(q)$.
- Each consumer obtains utility $u(q, \theta) - P_j(q)$ if purchasing from firm j.
- Cost: $C(q)$, increasing and convex.
- For any price schedule $P_j(q)$, there is an associated indirect utility $v_j(\theta) = \max_q [u(q, \theta) - P_j(q)]$. The schedule and the utility profile are two descriptions of the same object, so the firm may as well choose $v_j(\cdot)$ directly.
- Useful to include i.i.d. horizontal shocks, so that the "outside option" for j is $\underline{v}_j(\theta, \xi) = \max\{0, v_{-j}(\theta) - \xi_{-j}\} + \xi_j$, $\xi \sim G(\xi)$. These brand shocks are what keep the market from collapsing to Bertrand.

Competition in Indirect Utility Space

- Consumer θ chooses the firm offering the highest $v_j(\theta, \xi)$, so firm j's choice probability is $G_j(v_A(\theta), v_B(\theta))$.
- Substituting $P_j(q) = u(q(\theta), \theta) - v_j(\theta)$, the profit *per consumer of type* θ is

  $$\pi(v_j(\theta)) = u(q(\theta), \theta) - C(q(\theta)) - v_j(\theta)$$

  i.e. **total surplus generated minus the utility conceded**. Note $\pi'(v) = -1$: surplus is transferable one-for-one.
- Each firm maximizes over the *function* $v_j(\cdot)$:

  $$\max_{v_j} \; E_\theta \left[G_j(v_j(\theta), v_{-j}(\theta)) \, \pi(v_j(\theta)) \right] \quad \text{s.t.} \quad v_j'(\theta) = u_\theta(q(\theta), \theta)$$

  The single differential equation replaces the infinite set of global IC constraints — this is the same envelope trick as the continuum monopoly problem, and it is what makes the competitive problem tractable at all.
- This is an optimal-control problem (v the state, q the control). Rochet and Stole (2002) solve it; in general there is no closed form.

Symmetric Equilibrium with Full Coverage

- In symmetric equilibrium $v_A = v_B = v^*$. If the market is **fully covered**, every type buys from someone, so the only margin left is *which* firm — and that is governed by the horizontal shock ξ, not by θ. The firm's ability to screen on θ disappears.

- **First-order condition:**

$$\pi'(v^*)\, G(v^*, v^*) + \pi(v^*)\, G_1(v^*, v^*) = 0$$

Rearranging, and using $\pi' = -1$:

$$\frac{d \ln \pi(v)}{dv} = -\frac{d\sigma(v)}{dv}, \qquad \sigma(v) \equiv \ln G(v, v)$$

where σ measures the **intensity of competition** — how fast a firm's share responds to offering more utility.

Example: Logit Demand

- Suppose $G_A(v_A, v_B) = \dfrac{e^{v_A/\mu}}{e^{v_A/\mu} + e^{v_B/\mu}}$.
- Then $\dfrac{\partial G_A}{\partial v_A} = \dfrac{G_A(1 - G_A)}{\mu}$.
- Symmetric equilibrium $G_A = G_B = 1/2$ gives:

$$\frac{1}{2\mu}\pi(v^*) + \pi'(v^*)\frac{1}{2} = 0$$

- Thus equilibrium utility solves $\pi'(v^*) = -\dfrac{\pi(v^*)}{\mu}$, and with $\pi' = -1$ simply $\pi(v^*) = \mu$: **equilibrium profit per consumer equals the degree of horizontal differentiation**.
- Interpretation: higher μ (less intense competition) lets firms offer *less* utility, i.e. charge a higher fixed fee. As $\mu \to 0$ profit vanishes and consumers capture the entire surplus.

Equilibrium and Efficiency

- When firms are symmetric and all consumers buy, the equilibrium tariff is a **cost-based two-part tariff**: $P(q) = C(q) + F$, with F a fixed fee set by the intensity of competition.
- Marginal condition: $u_q(q, \theta) = C'(q)$ — the **efficient quantity for every type**, not just the top. Firms compete only via F (equivalently, via the level of v).

Result (Armstrong–Vickers 2001; Rochet–Stole 2002)

Under symmetry and full market coverage, competitive nonlinear pricing yields the **efficient allocation** — no distortion anywhere — and competition acts purely on the *transfer*.

- Why this benchmark matters empirically: it is sharply refutable, and McManus refutes it. Distortions survive in the coffee oligopoly, so full coverage must fail — consumers really do take the outside option, and firms screen against that margin.
- With partial coverage the solution interpolates between Mussa–Rosen and the cost-plus-fee benchmark: distortion smaller than monopoly, but not zero.

Three Designs, One Identification Problem

	Shepard (1991) gasoline	**Leslie (2004)** Broadway theater	**McManus (2007)** specialty coffee
Type of PD	2nd degree (service menu)	2nd + 3rd degree	2nd degree (size)
Screening device	full vs. self service	seat quality, queue at booth	cup size
Source of *mc*	*control group* of SP stations	assumed ≈ 0	*engineering data*
Structure needed	almost none	demand system + rationing	demand system
Identified from	configuration $\times$ service DiD	functional form + preset price calendar	ξ_j as fixed effects; cost data
Delivers	*existence* of PD	*welfare*: PD vs. uniform	*shape*: *MB* vs. *MC* by size
Headline	$\Delta \approx$ 9–11 c/gal; $\approx$ half the gap	PD -0.4% CS, $+5\%$ revenue	top efficient; small sizes $MB > 1.5\,MC$
Structure	local monopoly	monopoly (one show)	oligopoly

- The common thread: the real work is always a **marginal-cost benchmark not itself derived from the pricing behavior being tested**.

Additional Topics

Active areas in price discrimination we have not covered.

- **Bundling and multiproduct screening.** With multidimensional types the optimal mechanism is generally *not* a menu of quantities. Luo (2023) gives primitive conditions for when pure vs. mixed bundling is optimal, shows the structure is **nonparametrically identified**, and finds for China Telecom that mixed bundling of internet and landline benefits *both* firm and consumers. See also Crawford and Yurukoglu (2012) on cable.

- **Inter-temporal price discrimination.** Durable-goods pricing, clearance sales, advance purchase. Covered in the *Dynamic Demand* lecture.

- **Behavior-based and personalized pricing.** Pricing on purchase history or individual predicted willingness to pay (Dubé and Misra 2023). Algorithmic pricing and data regulation are where the field is most **active and least settled**.

Summary

- Bottom line for policy: no general presumption that price discrimination is harmful.
  - It redistributes — often *among* consumers more than from consumers to firms (Dubé–Misra; DellaVigna–Gentzkow), and banning it is not automatically progressive.
  - Whether it raises *total* surplus turns on whether it expands output: an estimation question, which is why this literature is structural once it moves past "does PD exist?"

- **Three open questions worth carrying forward:**
  1. Firms do not price optimally. Leslie and DellaVigna–Gentzkow both find clear departures from the model's optimum — inertia, fairness norms, industry convention. Our models assume otherwise; the gap is unexplained.
  2. The welfare criterion does hidden work. Once PD mostly reshuffles surplus *among* consumers, ΔCS is not a sufficient statistic; the answer depends on welfare weights we rarely state.
  3. Competition and screening interact in ways we have barely measured. Theory says full coverage removes the distortion and oligopoly is different; the empirical work is almost all monopoly.

References i

Theory and surveys

[1] Aguirre, I., S. Cowan, and J. Vickers (2010). "Monopoly Price Discrimination and Demand Curvature." *American Economic Review* 100(4), 1601–1615.

[2] Armstrong, M., and J. Vickers (2001). "Competitive Price Discrimination." *RAND Journal of Economics* 32(4), 579–605.

[3] Corts, K. S. (1998). "Third-Degree Price Discrimination in Oligopoly: All-Out Competition and Strategic Commitment." *RAND Journal of Economics* 29(2), 306–323.

[4] Holmes, T. J. (1989). "The Effects of Third-Degree Price Discrimination in Oligopoly." *American Economic Review* 79(1), 244–250.

References ii

[5] Mussa, M., and S. Rosen (1978). "Monopoly and Product Quality." *Journal of Economic Theory* 18(2), 301–317.

[6] Pigou, A. C. (1920). *The Economics of Welfare*. London: Macmillan.

[7] Rochet, J.-C., and L. A. Stole (2002). "Nonlinear Pricing with Random Participation." *Review of Economic Studies* 69(1), 277–311.

[8] Schmalensee, R. (1981). "Output and Welfare Implications of Monopolistic Third-Degree Price Discrimination." *American Economic Review* 71(1), 242–247.

[9] Stigler, G. J. (1987). *The Theory of Price*, 4th ed. New York: Macmillan.

[10] Stole, L. A. (2007). "Price Discrimination and Competition." In M. Armstrong and R. Porter (eds.), *Handbook of Industrial Organization*, Vol. 3, ch. 34, 2221–2299.

References iii

[11] Varian, H. R. (1985). "Price Discrimination and Social Welfare." *American Economic Review* 75(4), 870–875.

[12] Varian, H. R. (1989). "Price Discrimination." In R. Schmalensee and R. Willig (eds.), *Handbook of Industrial Organization*, Vol. 1, ch. 10, 597–654.

References iv

Empirical studies

[1] Crawford, G. S., and M. Shum (2007). "Monopoly Quality Degradation and Regulation in Cable Television." *Journal of Law and Economics* 50(1), 181–219.

[2] Crawford, G. S., O. Shcherbakov, and M. Shum (2019). "Quality Overprovision in Cable Television Markets." *American Economic Review* 109(3), 956–995.

[3] Crawford, G. S., and A. Yurukoglu (2012). "The Welfare Effects of Bundling in Multichannel Television Markets." *American Economic Review* 102(2), 643–685.

[4] DellaVigna, S., and M. Gentzkow (2019). "Uniform Pricing in US Retail Chains." *Quarterly Journal of Economics* 134(4), 2011–2084.

References v

[5] Dubé, J.-P., and S. Misra (2023). "Personalized Pricing and Consumer Welfare." *Journal of Political Economy* 131(1).

[6] Goldberg, P. K., and F. Verboven (2001). "The Evolution of Price Dispersion in the European Car Market." *Review of Economic Studies* 68(4), 811–848.

[7] Goldberg, P. K., and F. Verboven (2004). "Cross-Country Price Dispersion in the Euro Era: A Case Study of the European Car Market." *Economic Policy* 19(40), 483–521.

[8] Goldberg, P. K., and F. Verboven (2005). "Market Integration and Convergence to the Law of One Price: Evidence from the European Car Market." *Journal of International Economics* 65(1), 49–73.

References vi

[9] Grennan, M. (2013). "Price Discrimination and Bargaining: Empirical Evidence from Medical Devices." *American Economic Review* 103(1), 145–177.

[10] Gowrisankaran, G., A. Nevo, and R. Town (2015). "Mergers When Prices Are Negotiated: Evidence from the Hospital Industry." *American Economic Review* 105(1), 172–203.

[11] Leslie, P. (2004). "Price Discrimination in Broadway Theater." *RAND Journal of Economics* 35(3), 520–541.

[12] Luo, Y. (2023). "Bundling and Nonlinear Pricing in Telecommunications." *RAND Journal of Economics* 54(2), 268–298.

[13] McManus, B. (2007). "Nonlinear Pricing in an Oligopoly Market: The Case of Specialty Coffee." *RAND Journal of Economics* 38(2), 512–532.

[14] Shepard, A. (1991). "Price Discrimination and Retail Configuration." *Journal of Political Economy* 99(1), 30–53.